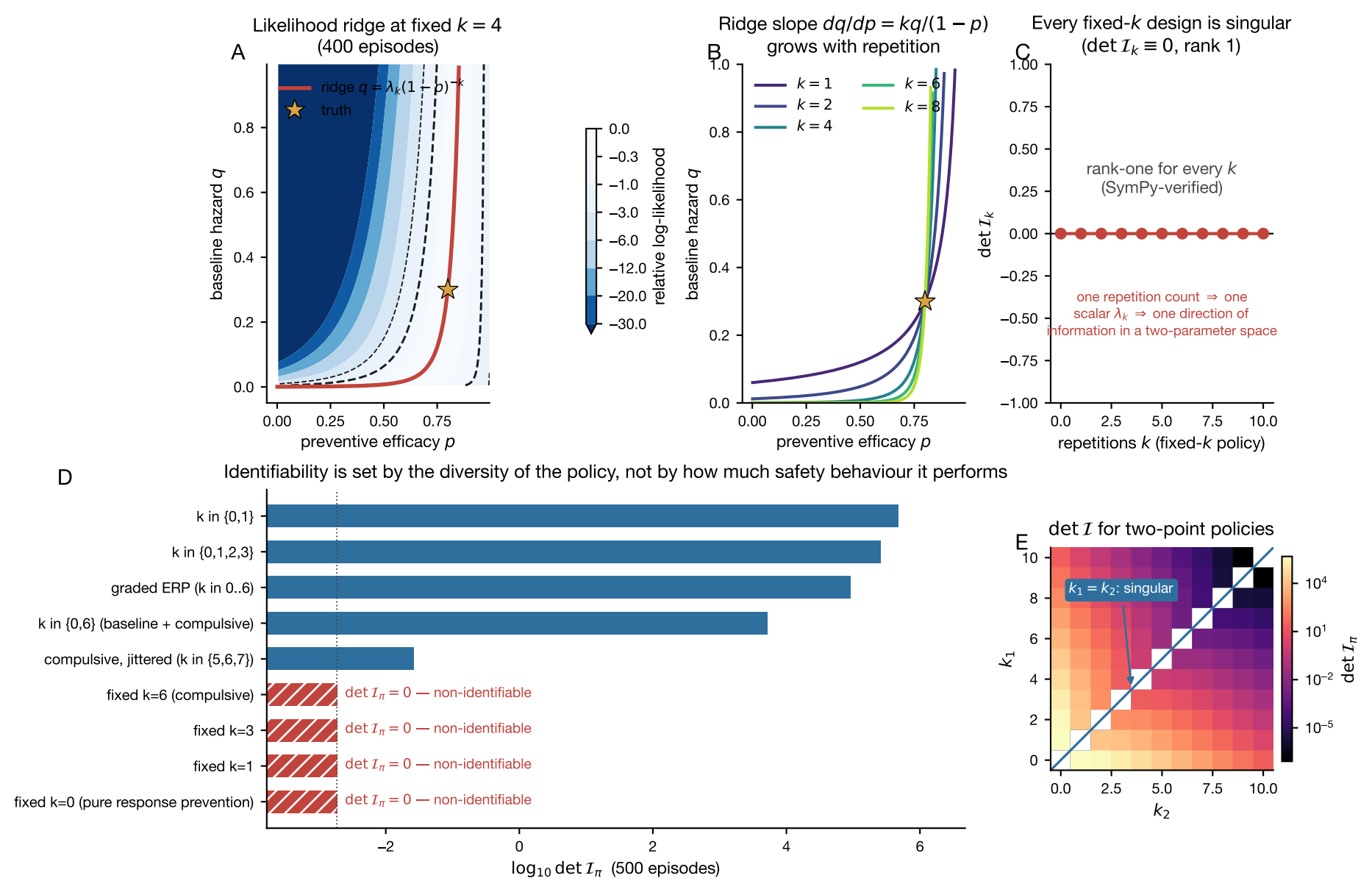

A
Likelihood ridge at fixed $k = 4$
(400 episodes)
ridge $q = \lambda_k(1-p)^{-k}$
truth
baseline hazard q
preventive efficacy p
relative log-likelihood
0.0
−0.3
−1.0
−3.0
−6.0
−12.0
−20.0
−30.0
B
Ridge slope $dq/dp = kq/(1-p)$
grows with repetition
$k = 1$
$k = 2$
$k = 4$
$k = 6$
$k = 8$
baseline hazard q
preventive efficacy p
C
Every fixed-k design is singular
(det $\mathcal{I}_k \equiv 0$, rank 1)
rank-one for every k
(SymPy-verified)
one repetition count ⇒ one
scalar λ_k ⇒ one direction of
information in a two-parameter space
det $\mathcal{I}_k$
repetitions k (fixed-k policy)
D
Identifiability is set by the diversity of the policy, not by how much safety behaviour it performs
k in {0,1}
k in {0,1,2,3}
graded ERP (k in 0..6)
k in {0,6} (baseline + compulsive)
compulsive, jittered (k in {5,6,7})
fixed k=6 (compulsive)
fixed k=3
fixed k=1
fixed k=0 (pure response prevention)
det $\mathcal{I}_\pi = 0$ — non-identifiable
$\log_{10}$ det $\mathcal{I}_\pi$ (500 episodes)
E
det $\mathcal{I}$ for two-point policies
$k_1 = k_2$: singular
k_1
k_2
det $\mathcal{I}_\pi$